

Let D be the set of detectors in the benchmark. For $d \in D$, define three raw features from its reference implementation:

$$x_1(d) = \text{SLOC}(d), \quad x_2(d) = \text{NumScanners}(d), \quad x_3(d) = \text{NumTestMethods}(d)$$

Let $m_k = \text{median}\{x_k(d) : d \in D\}$ for $k \in \{1, 2, 3\}$, computed once over the whole benchmark.

Define the per-axis indicator:

$$\mathbb{1}_k(d) = \begin{cases} 1 & \text{if } x_k(d) > m_k \\ 0 & \text{otherwise} \end{cases}$$

Raw score:

$$s(d) = \sum_{k=1}^3 \mathbb{1}_k(d) \in \{0, 1, 2, 3\}$$

Normalized difficulty score:

$$\text{difficulty_score}(d) = \frac{s(d)}{3} \in \{0, \frac{1}{3}, \frac{2}{3}, 1\}$$

Tier:

$$\text{tier}(d) = \begin{cases} \text{EASY} & s(d) = 0 \\ \text{MEDIUM} & s(d) \in \{1, 2\} \\ \text{HARD} & s(d) = 3 \end{cases}$$

Every instance i generated from detector d inherits $\text{tier}(d)$.